Apparent Catamenial Epilepsy Classification Under a Null Simulation

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Abstract

Objective: To quantify apparent catamenial epilepsy classification under a null model in which seizure and menstrual-cycle diaries are independent, with emphasis on C1, C2, and C3 pattern-specific false positives. Methods: We simulated 100,000 synthetic participants for 36 months in healthy ovulatory and heterogeneous menstruating-age cohorts. CHOCOLATES generated seizure diaries and HORMONE-CYCLE generated independent menstrual/hormone diaries. A random circular shift was applied before merging. Strict Herzog phase labels were primary, with a luteal-anchored fixed periovulatory sensitivity. We evaluated exact Herzog 2004, windowed Herzog thresholds, Herzog plus >=4 seizure-day minimums, and cycle reproducibility >=2/3 across calendar, cycle, and full-diary windows; negative-binomial C1/C2 regression was evaluated as a full-diary calibrated comparator. Results: In 36-month strict-Herzog windows, windowed Herzog thresholds classified 11.2% of healthy ovulatory and 36.3% of heterogeneous-cohort participants as apparent catamenial epilepsy under independence. In the heterogeneous cohort, C3 drove the excess: the C1/C2 union was 11.8%, whereas C3 positivity occurred in 37.0% of applicable windows. Negative-binomial C1/C2 regression exhibited approximately nominal Type I error, 4.1% and 4.2%. Three-month calendar windows remained vulnerable, with windowed Herzog false-positive rates of 41.3% and 51.2%. Conclusions: Apparent catamenial epilepsy classification can be common under independence, especially when C3/inadequate-luteal-phase logic is applied in heterogeneous cycles. Studies should report C1, C2, and C3 separately and prespecify diary duration, seizure-day minimums, denominator handling, phase labeling, and reproducibility criteria.

Classification of Evidence: Not applicable. This simulation study evaluates operating characteristics of classification rules under a synthetic null model and does not test a diagnostic intervention or therapeutic effect in human participants.

Search Terms: catamenial epilepsy; epilepsy; seizures; menstrual cycle; simulation study.

Introduction

Catamenial epilepsy describes seizure exacerbation in relation to the menstrual cycle. The clinical literature has long recognized perimenstrual, periovulatory, and inadequate luteal-phase patterns, but reported prevalence varies widely because studies use different phase definitions, thresholds, diary durations, and eligibility rules. Herzog and colleagues reported 39.1% catamenial epilepsy in women who charted seizures and menses for three complete cycles using a two-of-three-cycle rule, and later work from the NIH Progesterone Treatment Trial has been interpreted as supporting prevalence near 44% in selected trial participants.1,5

The diagnostic problem is not only biological. Seizures are temporally structured: clustering, circadian rhythms, and multidien rhythms have been documented in long-term diaries and intracranial recordings.9-11 If seizure rhythms are independent of menstrual cycles but contain multidien structure in overlapping period bands, finite diary windows can still contain chance alignments. Ratio definitions are especially vulnerable when seizure counts are sparse or comparator phases contain few events. This creates a specificity question with direct implications for clinical research: what apparent catamenial epilepsy rate should be expected when no menstrual coupling exists?

We conducted a reproducible null-simulation study to estimate apparent catamenial epilepsy classification across definitions and observation windows under independence. The analysis evaluates specificity-like behavior only. Because traditional definitions combine C1, C2, and C3 patterns, we separately report C1/C2 and C3 false-positive behavior and compare negative-binomial regression only against C1/C2 outcomes.

Methods

Study design. This Monte Carlo null simulation was defined before the full simulation run. For each synthetic participant, seizure and menstrual/hormone diaries were generated independently for 36 months. A participant-specific random circular shift of the seizure diary was sampled uniformly over the diary length before merging to prevent hidden start-date alignment; cycle labels were preserved from the hormone diary and seizure counts were wrapped with the shifted seizure diary. Random calendar and cycle windows were sampled per participant and could overlap across analysis families; all analyses were stratified by cohort and phase-labeling mode.

Cohorts. The healthy ovulatory cohort included 50,000 synthetic participants, age 18-45 years, with ovulatory cycling enforced through the simulator adapter where available. The heterogeneous menstruating-age cohort included 50,000 synthetic participants, age 13.0-54.9 years, allowing anovulation and irregularity. Because the hormone-cycle simulator exposed medical-factor controls but not a natural prevalence sampler, PCOS, peri-menarche, perimenopause, and dysmenorrhea were sampled from configuration rates. This cohort should be interpreted as an assumption-driven heterogeneity stress test unless these rates are replaced with externally validated prevalence inputs.

Diary simulators. CHOCOLATES (Cyclic Heterogeneous Overdispersed Clustered Open-source L-relationship Adjustable Temporally limited E-diary Simulator) generated non-catamenial seizure diaries using a hierarchical gamma-Poisson/negative-binomial seizure-count generator with seizure-burden heterogeneity, multidien cycles, clustering, mean-variance structure, and interseizure-interval limits.[12] HORMONE-CYCLE generated independent menstrual/hormone diaries. It samples participant-level latent traits and cycle-level realizations, then interpolates daily estradiol and progesterone trajectories from menstrual subphase reference medians. Its features include cycle length, within-participant variability, bleeding days, ovulatory and anovulatory cycles, follicular and luteal phase lengths, ovulation day, luteal progesterone, inadequate-luteal-phase flags, and configured modifiers for PCOS, peri-menarche, perimenopause, dysmenorrhea, oral contraceptive use, hormonal IUD, and copper IUD. Baseline cycle targets were based on Apple Women's Health Study data, Natural Cycles data, and Stricker et al. hormone reference values.13-15 Simulator validation targets, versions, and package commits are documented in the Appendix following STRESS reporting principles.16

Phase labeling and definitions. Strict Herzog phases were primary. In a complete cycle of length L, day 1 was menstrual-flow onset and backward day was d-(L+1). Strict labels were assigned as menstrual phase if forward day 1-3 or backward day -3 to -1; follicular phase if forward day 4-9; ovulatory phase if forward day >=10 and backward day <=-13; luteal phase if backward day -12 to -4; otherwise unlabeled. Because this strict ovulatory window expands with cycle length, we added a sensitivity mode that fixed the ovulatory phase at backward days -16 to -13 immediately before the luteal phase, with the follicular phase absorbing cycle-length variability.

Windowed Herzog thresholds used average daily seizure frequency (ADSF): C1 = ADSF(M)/ADSF(F+L) >=1.69, C2 = ADSF(O)/ADSF(F+L) >=1.83, and C3 = ADSF(O+L+M)/ADSF(F) >=1.62 among inadequate-luteal-phase days. C3 was evaluated only in the heterogeneous cohort when ilp_flag was present; anovulatory or nonclassifiable cycles without ilp_flag did not contribute to C3. We report any C1-C3 classification, the C1/C2 union, mutually exclusive C1/C2/C3 categories, and C3-excluding endpoints.

Exact Herzog 2004 was restricted to complete 3-cycle windows with all cycles 23-35 days and required at least two of three cycles to show any positive pattern. Other calendar and longer cycle windows are reported as not evaluated for this rule. Herzog plus >=4 seizure-day minimum required at least 4 calendar months or 6 complete cycles plus at least 4 seizure days. Cycle reproducibility required at least two-thirds of eligible cycles to be positive for the same pattern and the corresponding pooled ratio to pass threshold.

Regression comparator. The negative-binomial (NB) comparator used a within-participant generalized linear model with daily seizure count as outcome, log link, intercept, menstrual-phase indicator, ovulatory-phase indicator, and cycle fixed effects when at least four complete cycles were available. One-sided Wald tests evaluated C1 and C2 enrichment, Holm-adjusted within participant across the two comparisons; positivity required adjusted $P<0.05$ and the corresponding Herzog rate-ratio threshold. C3 was not evaluated by NB regression. In this revision, NB regression was evaluated for full-diary windows as a calibrated C1/C2 comparator. The primary regression used full-diary method-of-moments dispersion as a stabilized comparator, and a full-window, window-only dispersion sensitivity was added.

Statistical analysis. The primary outcome was the participant-window apparent catamenial epilepsy rate among classifiable windows under the null. We also report positivity among all attempted windows, participant-level positivity, indeterminate reasons, and rows with fewer than 1,000 classifiable windows as unstable and not interpreted. Calendar windows were expanded to 1, 3, 4, 6, 9, 12, 18, 24, and 36 months for practical monitoring-duration guidance. Study-level Monte Carlo independently sampled 10,000 studies of 30, 50, and 100 participants without replacement from each cohort, selecting one precomputed random valid 3-month window per participant. Standard protocol approvals, registrations, and patient consents were not required because all participants were synthetic.

Results

Simulation cohort and diary characteristics. The full run included 100,000 synthetic participants. With strict and luteal-anchored phase modes, expanded calendar windows, cycle windows, and study-pool windows, the analysis generated paired participant-window rows for sensitivity comparisons. Cohort summaries are shown in Table 1. The healthy ovulatory cohort had 100.0% ovulatory cycles by design. The heterogeneous cohort had 79.0% ovulatory cycles, greater cycle-length variability, and similar observed seizure burden.

C3 contribution and C1/C2 comparison. In strict-Herzog 36-month windows, the windowed Herzog threshold definition classified 11.2% of healthy ovulatory participants and 36.3% of heterogeneous-cohort participants as apparent catamenial epilepsy under independence (Table 2). In the heterogeneous cohort, the corresponding C1/C2 union was 11.8%, similar to the healthy cohort's 11.2%, whereas C3 positivity occurred in 37.0% of applicable heterogeneous-cohort windows. Thus the broad-cohort excess was primarily a C3/inadequate-luteal-phase phenomenon rather than a perimenstrual or periovulatory excess.

NB regression and denominator alignment. Stabilized-dispersion NB regression evaluated C1/C2 only and exhibited approximately nominal Type I error at 4.1% in the healthy ovulatory cohort and 4.2% in the heterogeneous cohort. The relevant comparison is therefore NB C1/C2 versus Herzog C1/C2, not NB versus the Herzog C1-C3 union. Calibration should not be interpreted as clinical usefulness; sensitivity and positive predictive value require true-coupling analyses outside this null study.

Observation-window effects. Short diary windows were most vulnerable to apparent catamenial classification. In random 3-month calendar windows, strict-Herzog windowed thresholds yielded apparent classification rates of 41.3% in the healthy ovulatory cohort and 51.2% in the heterogeneous cohort. Practical calendar-window summaries through 36 months are shown in Figure 1. Exact Herzog 2004 applied to 3 complete cycles yielded apparent classification rates of 50.3% and 51.6% among classifiable windows, while many attempted 3-cycle windows were not classifiable. Rows with fewer than 1,000 classifiable windows were flagged as unstable and not interpreted.

Phase-labeling sensitivity and study-level Monte Carlo. The luteal-anchored ovulatory sensitivity tested whether strict Herzog's expanding ovulatory window drove periovulatory false positives; these results are reported in the notebook and Appendix. In 10,000 simulated 3-month studies, apparent prevalence varied with study size but could exceed historical benchmark values under the null. Figure 3 displays n=30, 50, and 100 distributions with the y-axis scaled as the proportion of simulated studies.

Discussion

This null simulation shows that apparent catamenial epilepsy can arise frequently even when seizure and menstrual processes are generated independently. The magnitude depends strongly on the catamenial pattern being tested, the window length, analyzability rules, and cohort menstrual-cycle assumptions. The largest heterogeneous-cohort excess was C3-driven; C1/C2 behavior was much more similar across cohorts. This distinction matters because C3 depends on inadequate-luteal-phase ascertainment rather than the perimenstrual or periovulatory windows that are usually easiest to audit from diaries alone.

The negative-binomial result should be interpreted as calibration, not superiority. A statistical test evaluated at alpha=0.05 should produce approximately 5% positives under a correctly specified null, and the stabilized-dispersion regression did so. Ratio rules are not nominal-alpha tests and therefore have no comparable Type I error guarantee. The relevant comparison in this paper is NB C1/C2 versus the Herzog C1/C2 union, because the NB model does not evaluate C3. Sensitivity, positive predictive value, and clinical usefulness require true-coupling analyses and are outside the scope of this null study.

These results help interpret the broad range of prevalence estimates in the catamenial epilepsy literature without estimating true prevalence. Earlier work established biologically plausible and clinically important menstrual-cycle seizure patterns, including C1, C2, and C3 patterns.[1-5] Treatment literature also suggests that robust perimenstrual exacerbation may identify subgroups of interest, even though randomized evidence for hormonal therapy remains limited and mixed.[6,7] The present analysis instead quantifies a diagnostic background rate expected from chance alignment when seizure cycles and menstrual cycles coexist under independence.

The findings are consistent with modern seizure-cycle research. Multidien seizure rhythms and seizure forecasting studies have shown that seizure risk varies over days to weeks, and those rhythms can be detected in long-term electronic diaries and chronic recordings.[9-11] A menstrual-cycle analysis that does not account for intrinsic seizure cyclicity can therefore mistake phase overlap for hormone coupling. This is most problematic when the observation window is short, because a few events can dominate average daily seizure-frequency ratios.

Several methodologic implications follow. First, 3-month ratio-only classification should be interpreted cautiously in research settings, particularly when seizure counts are sparse. Second, C1, C2, and C3 should be reported separately rather than collapsed into a single prevalence value. Third, studies should report both classifiable-window and all-attempted-window rates, because stricter definitions may appear conservative partly by declining to classify insufficient windows. Fourth, benchmark comparisons should match the original study design. For Herzog

2004, the design-matched comparison is the exact 3-complete-cycle rule, whereas random 3-calendar-month windows are a design-sensitivity analysis.

Strict Herzog labeling was retained as the primary analysis for historical comparability. The luteal-anchored ovulatory sensitivity addresses a separate biologic question: whether the strict Herzog ovulatory window, which expands with longer cycles, creates excess periovulatory apparent classification. Presenting both phase modes separates historical reproducibility from a more anatomically fixed periovulatory definition.

This study has limitations. The results are only as credible as the CHOCOLATES and HORMONE-CYCLE simulators, their parameter settings, and the adapter assumptions. The heterogeneous cohort medical-factor rates were configuration-sampled rather than drawn from a simulator-native prevalence sampler, and cycle-length variability is therefore a sensitivity driver rather than a settled population estimate. Monte Carlo binomial intervals do not include model-form uncertainty. This study does not evaluate false negatives, true prevalence, treatment effects, wavelet analyses, hidden Markov models, medication changes, or adjudicated seizure types.

In conclusion, under an independence model with structured seizure diaries and menstrual-cycle diaries, apparent catamenial epilepsy classification can be common and can reach historical benchmark prevalence values. Future diagnostic and interventional studies should prespecify diary length, menstrual-phase labeling, analyzability thresholds, zero-denominator handling, ovulatory-status handling, and reproducibility criteria before interpreting apparent menstrual clustering as evidence of hormone-linked seizure exacerbation.

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Disclosure

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Data Availability

The simulation code, configuration, analysis outputs, and machine-readable manifest are available in the project repository. The reproducible run used master seed 20260505 and the manifest in outputs/manifest.json records config parameters, output paths, file sizes, and SHA-256 checksums. A versioned public archive with DOI will accompany the submitted version.

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Tables

Table 1. Simulated cohort characteristics.

Cohort	N	Mean age, y	Mean cycle length, d	Cycle-length SD, d	Ovulatory cycles	Seizure days/month	Seizures/ month
healthy ovulatory	50,000	31.5 (SD 7.8)	29.15	3.39	100.0%	2.46	6.83
heterogeneous menstruating-age	50,000	34.0 (SD 12.1)	30.92	4.74	79.0%	2.45	6.80

Values are cohort-level means from the full 100,000-participant run. Cycle-length SD is the mean within-participant cycle-length SD.

Table 2. Participant-window false-positive results.

Cohort	Analysis setting	Definition	Classifiable / attempted	Positive among classifiable	Positive among all attempted	Indeterminate
healthy ovulatory	36-month full diary	Windowed Herzog thresholds	49,605 / 50,000	11.2% (11.0%-11.5%)	11.2%	0.8%
healthy ovulatory	36-month full diary	Windowed Herzog C1/C2 union	49,605 / 50,000	11.2% (11.0%-11.5%)	11.2%	0.8%
healthy ovulatory	36-month full diary	Windowed Herzog C3 only	0 / 50,000	NA	0.0%	100.0%
healthy ovulatory	36-month full diary	Herzog C1/C2 plus >=4 seizure-day minimum	48,359 / 50,000	10.2% (9.9%-10.5%)	9.9%	3.3%
healthy ovulatory	36-month full diary	Negative-binomial regression C1/C2	48,359 / 50,000	4.1% (3.9%-4.2%)	3.9%	3.3%
healthy ovulatory	3-month calendar window	Windowed Herzog thresholds	45,280 / 50,000	41.3% (40.8%-41.7%)	37.4%	9.4%
healthy ovulatory	3 complete cycles	Exact Herzog 2004, any pattern	22,996 / 50,000	49.7% (49.0%-50.3%)	22.8%	54.0%
heterogeneous menstruating-age	36-month full diary	Windowed Herzog thresholds	49,587 / 50,000	36.3% (35.8%-36.7%)	36.0%	0.8%

Cohort	Analysis setting	Definition	Classifiable / attempted	Positive among classifiable	Positive among all attempted	Indeterminate
heterogeneous menstruating-age	36-month full diary	Windowed Herzog C1/C2 union	49,587 / 50,000	11.8% (11.5%-12.1%)	11.7%	0.8%
heterogeneous menstruating-age	36-month full diary	Windowed Herzog C3 only	38,989 / 50,000	37.0% (36.5%-37.5%)	28.8%	22.0%
heterogeneous menstruating-age	36-month full diary	Herzog C1/C2 plus >=4 seizure-day minimum	48,333 / 50,000	10.7% (10.5%-11.0%)	10.4%	3.3%
heterogeneous menstruating-age	36-month full diary	Negative-binomial regression C1/C2	48,333 / 50,000	4.2% (4.0%-4.4%)	4.1%	3.3%
heterogeneous menstruating-age	3-month calendar window	Windowed Herzog thresholds	45,297 / 50,000	51.2% (50.7%-51.6%)	46.4%	9.4%
heterogeneous menstruating-age	3 complete cycles	Exact Herzog 2004, any pattern	16,757 / 50,000	51.2% (50.4%-51.9%)	17.2%	66.5%

All rows use strict Herzog phase labeling. Rows with fewer than 1,000 classifiable windows are flagged in the analysis outputs and not interpreted.

Table 3. C1/C2/C3 mutually exclusive pattern decomposition.

Cohort	Definition	C1 only	C2 only	C1+C2	C3 only	C3 plus C1/C2	None	Indeterminate
healthy ovulatory	Windowed Herzog thresholds	5.6%	3.4%	2.1%	0.0%	0.0%	88.1%	0.8%
healthy ovulatory	Herzog plus >=4 seizure-day minimum	5.1%	2.9%	1.9%	0.0%	0.0%	86.8%	3.3%
heterogeneous menstruating-age	Windowed Herzog thresholds	3.8%	2.1%	1.2%	24.3%	4.6%	63.2%	0.8%
heterogeneous menstruating-age	Herzog plus >=4 seizure-day minimum	3.5%	1.7%	1.0%	24.0%	4.2%	62.3%	3.3%

Percentages are among all attempted 36-month full-diary windows, so indeterminate windows remain visible in the denominator.

Table 4. Study-level Monte Carlo summaries for 3-month windows.

Cohort	**N per study**	**Definition**	**Mean apparent prevalence**	**2.5th-97.5th percentiles**	**P >=39.1%**	**P >=44.2%**
healthy ovulatory	30	Windowed Herzog thresholds	37.5%	20.0%-53.3%	45.4%	20.1%
healthy ovulatory	30	Windowed Herzog C1/C2 union	37.5%	20.0%-53.3%	45.4%	20.1%
healthy ovulatory	50	Windowed Herzog thresholds	37.3%	24.0%-50.0%	39.7%	13.0%
healthy ovulatory	50	Windowed Herzog C1/C2 union	37.3%	24.0%-50.0%	39.7%	13.0%
healthy ovulatory	100	Windowed Herzog thresholds	37.5%	28.0%-47.0%	33.8%	7.1%
healthy ovulatory	100	Windowed Herzog C1/C2 union	37.5%	28.0%-47.0%	33.8%	7.1%
heterogeneous menstruating-age	30	Windowed Herzog thresholds	46.3%	30.0%-63.3%	80.5%	55.9%
heterogeneous menstruating-age	30	Windowed Herzog C1/C2 union	38.2%	20.0%-56.7%	48.9%	22.4%
heterogeneous menstruating-age	50	Windowed Herzog thresholds	46.1%	32.0%-60.0%	84.5%	55.9%
heterogeneous menstruating-age	50	Windowed Herzog C1/C2 union	38.1%	26.0%-52.0%	43.9%	15.3%
heterogeneous menstruating-age	100	Windowed Herzog thresholds	46.0%	36.0%-56.0%	90.8%	62.2%
heterogeneous menstruating-age	100	Windowed Herzog C1/C2 union	38.0%	29.0%-47.0%	38.2%	8.8%

Study-level rows summarize 10,000 simulated studies per cohort and sample size using one random valid 3-month window per selected participant. N=30 is illustrative; N=50 and N=100 are sensitivity rows.

Figure Legends

Figure 1. False-positive rates by observation window.

Bars show the percentage of classifiable calendar-window participant windows classified as catamenial epilepsy under the null using strict Herzog phase labeling. Dashed and dotted horizontal lines mark 5% and 10% reference rates.

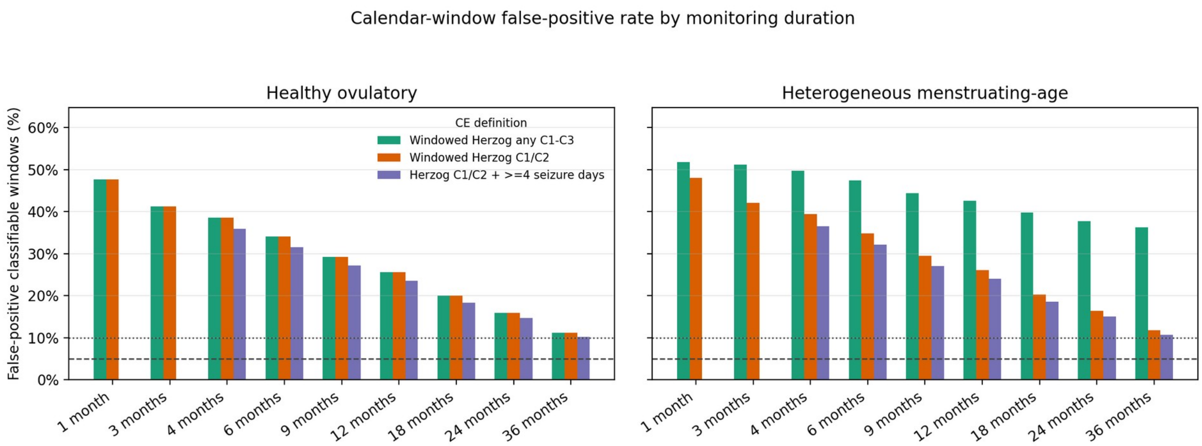

Figure 2. C1/C2/C3 mutually exclusive pattern decomposition.

Stacked bars show mutually exclusive pattern categories for 36-month full-diary windows by cohort and definition. Percentages use all attempted participant windows as the denominator.

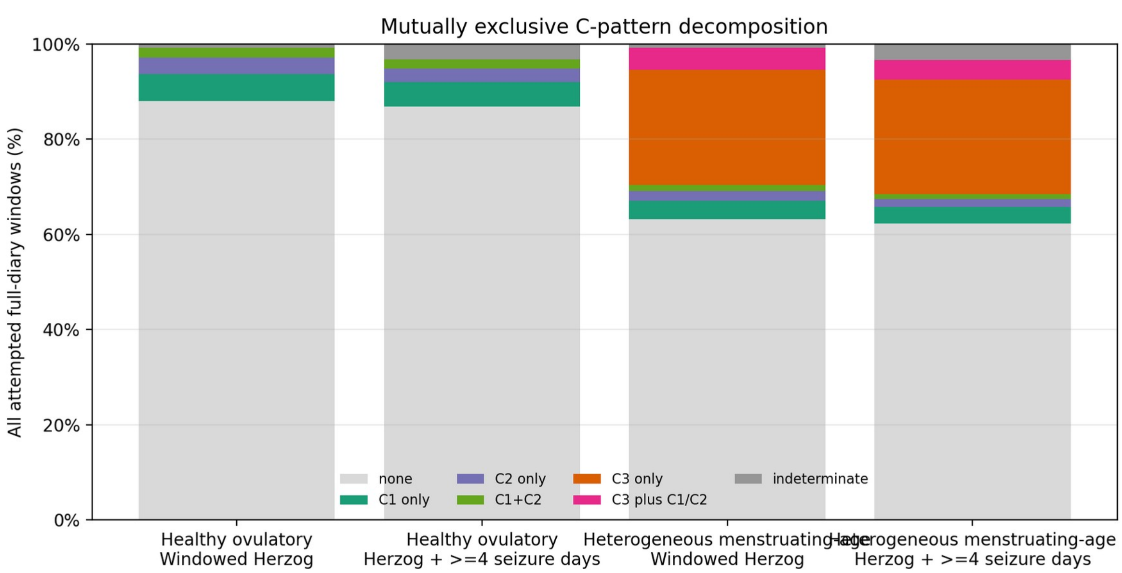

Figure 3. Apparent prevalence in null studies.

Distributions show apparent catamenial epilepsy prevalence in 10,000 simulated 3-month studies using strict Herzog windowed thresholds. Curves are shown for n=30, n=50, and n=100 participants; vertical lines show benchmark prevalence values of 39.1% and 44.2%.